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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SSH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SSH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtmpadr proto kernel_r
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:            0.0
Usage of /:             5.1% of 48.91GB
Memory usage:          5%
Swap usage:            0%
Processes:             139
Users logged in:       1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Course/Year/Section: BSIT3B
Date Submitted: 02/09/2026
Instructor: Puyo

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☐ Create and configure a Virtual Machine with Ubuntu Server
- ☐ Configure Bridged Network Adapter for network access
- ☐ Determine and verify the server's IP address
- ☐ Install OpenSSH Server and enable the service
- ☐ Establish successful SSH connection from host machine
- ☐ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Linux
Virtualization Software (VirtualBox/VMware)	oracle
Ubuntu Server Version	Ubuntu 64bit
VM Name	ROJO-ELECTIVE-3B
RAM Allocated	4
Storage Allocated	100GB
Network Mode	Intel PRO
SSH Client Used	Secure Shell Client

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	10240 MB
Hard Disk	100 GB

Network Adapter	Bridged
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4. NETWORK CONFIGURATION

Commands Executed: ssh joaquim@192.168.56.101

Server IP Address: 192.168.56.101

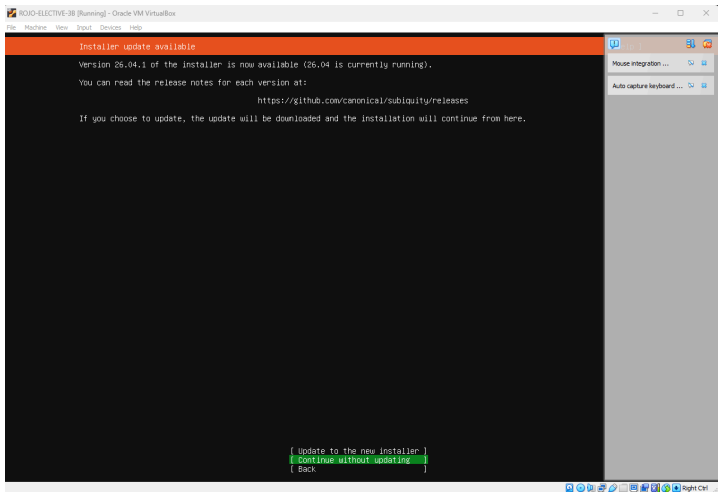
Network Interface: Host only adapter

6. REMOTE CONNECTION

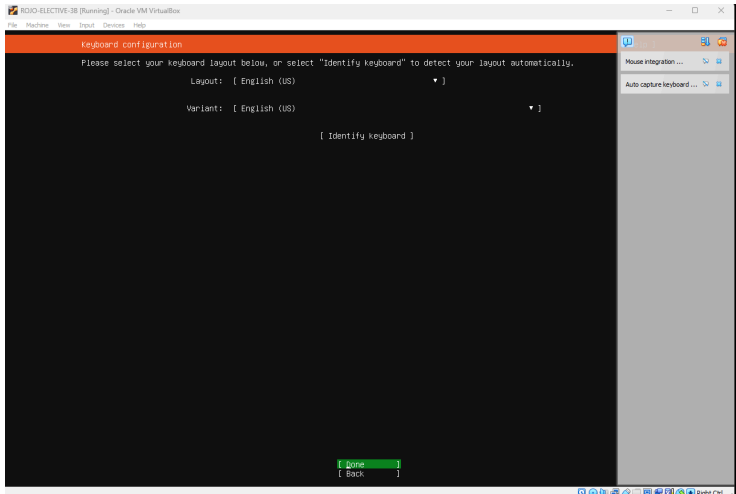
Connection Details:

Parameter	Value
Server IP	192.168.56.101
Username	joaquim
SSH Port	22

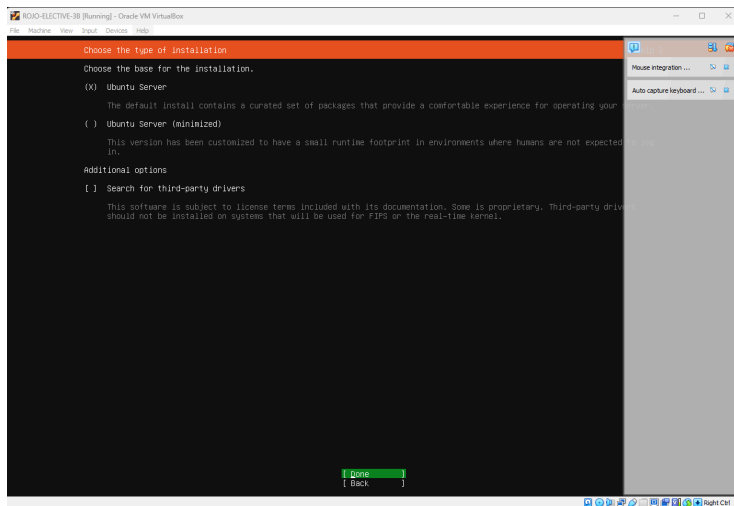
Select language



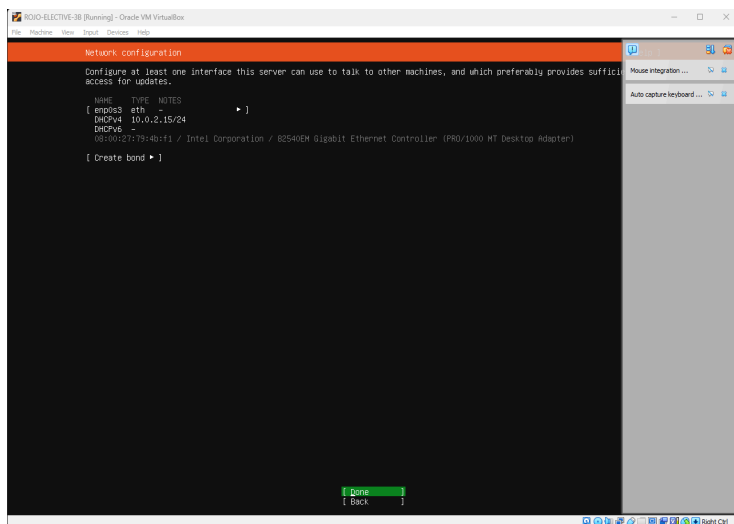
Continue without updating to proceed



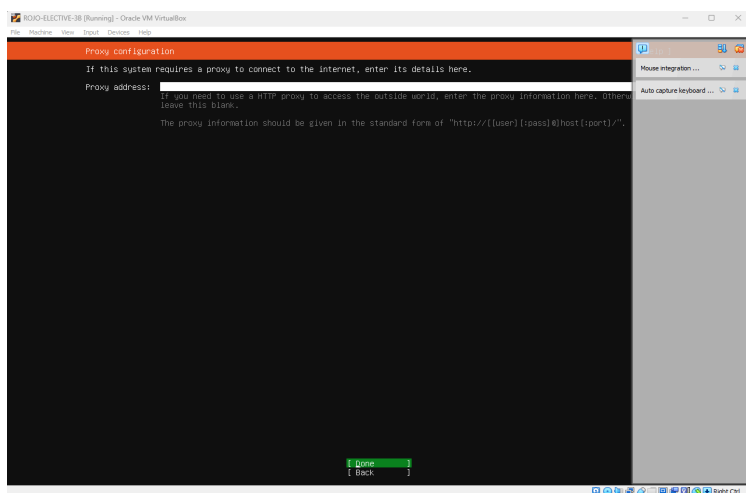
Confirm language



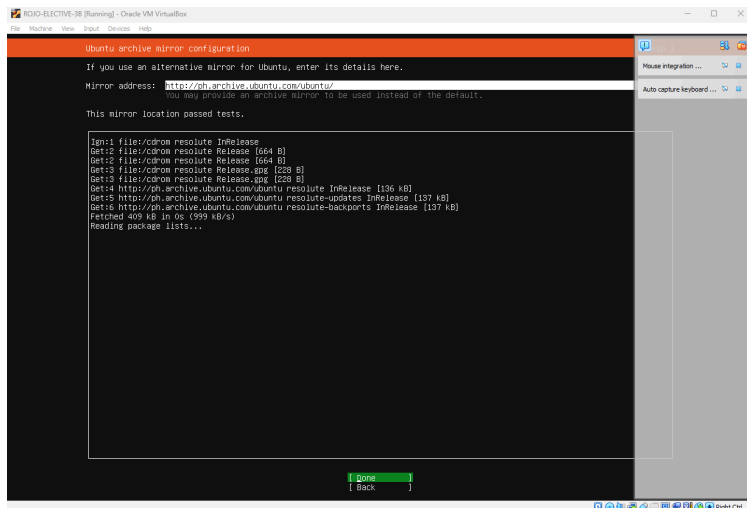
Download ubuntu server



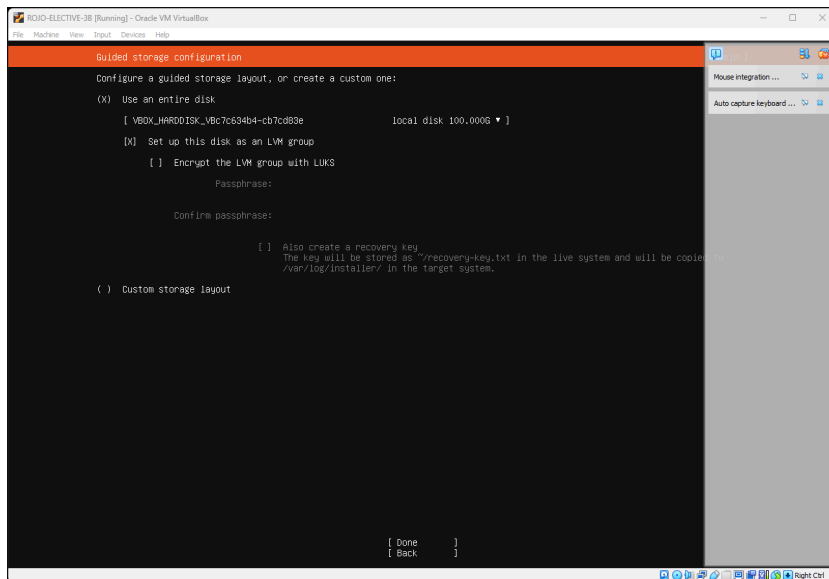
Confirm



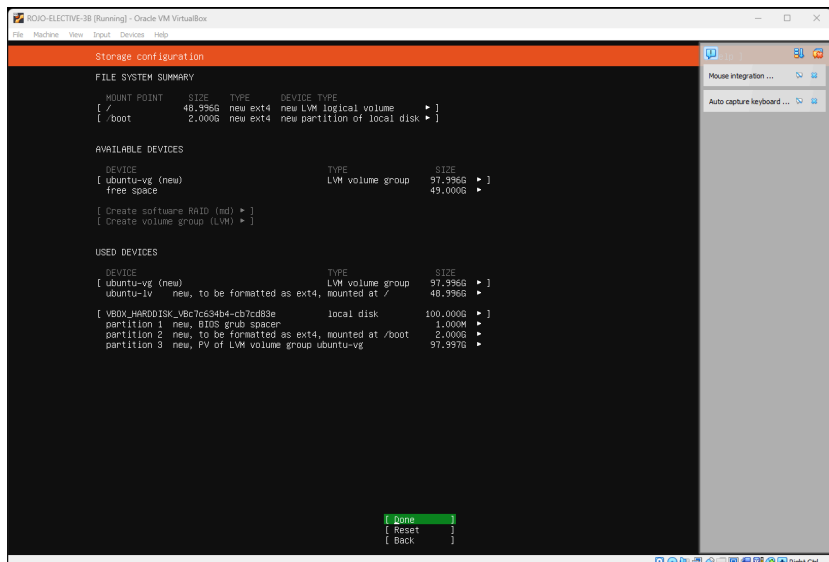
Continue



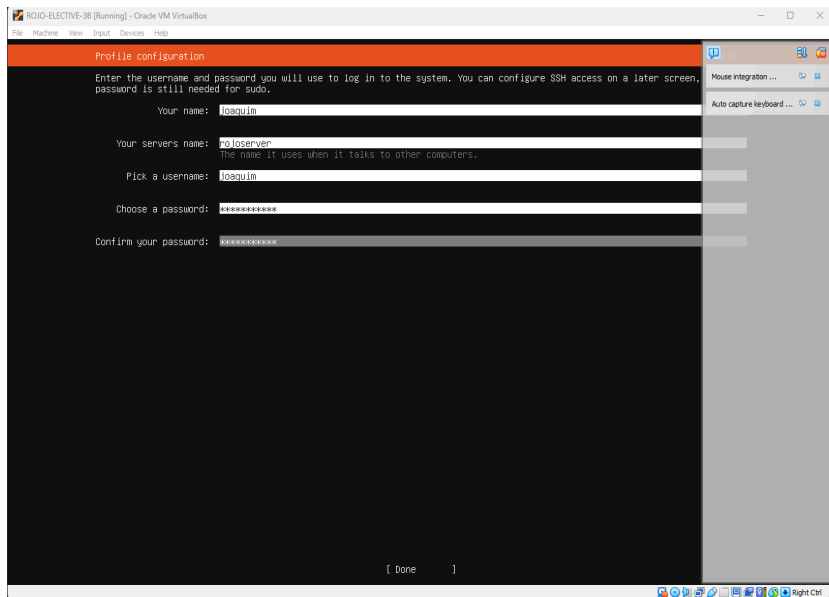
Wait to download then continue



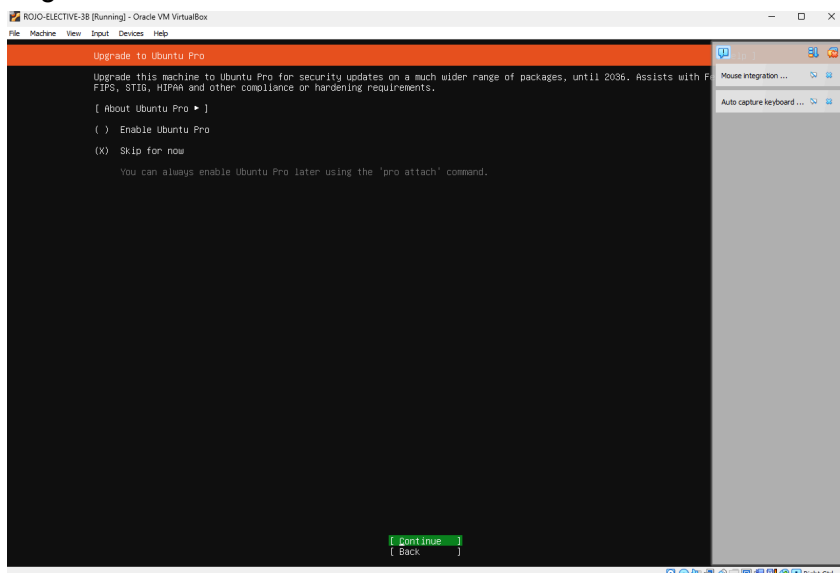
Use an entire disk



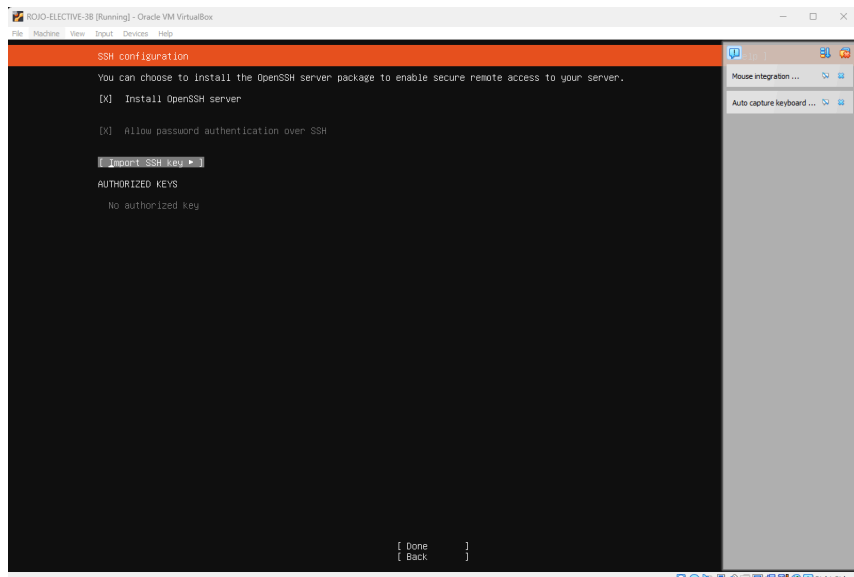
Continue



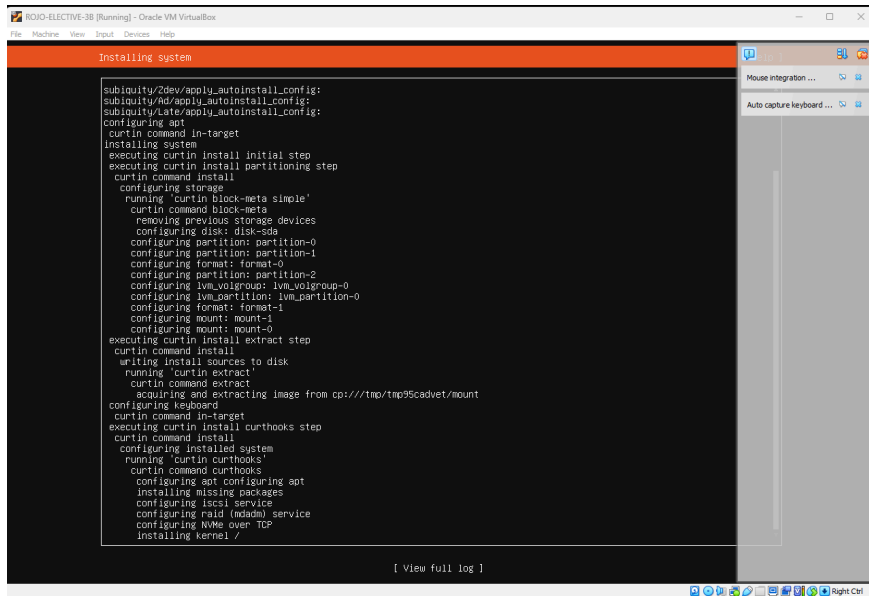
Register



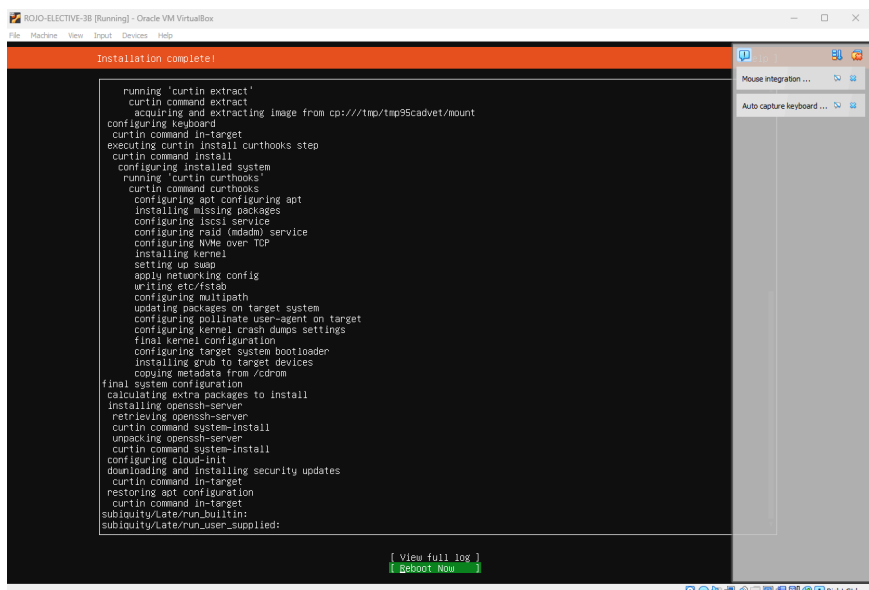
Skip to continue



Import SSH key



Wait to download



Reboot

```
ROID-ELECTIVE-38 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

ES updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

joaquim@rojo-server:~$ sudo systemctl status ssh
(sudo: authenticate) Password:
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Wed 2025-09-02 06:25:41 UTC; 2min 34s ago
   Invocation: 6cc373de6b046f94ef8ef8eb1dc826
   Process: ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Process: 1395 ExecStart=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1395 (sshd)
   Tasks: 1 (limit: 8418)
   Memory: 1.0M (peak: 2.5M)
   CPU: 18ms
   CGroup: /system.slice/ssh.service
           └─1395 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 02 06:25:41 rojo-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 02 06:25:41 rojo-server sshd[1395]: Server listening on 0.0.0.0 port 22.
Sep 02 06:25:41 rojo-server sshd[1395]: Server listening on :: port 22.
Sep 02 06:25:41 rojo-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
joaquim@rojo-server:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/ssh.service → /usr/lib/systemd/system/ssh.service'.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service'.
joaquim@rojo-server:~$ sudo systemctl start ssh
Failed to start ssh.service: Unit ssh.service not found.
joaquim@rojo-server:~$ sudo apt update
Hit:1 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
joaquim@rojo-server:~$ sudo apt install openssh-server-y
```

Login to your account

```
ROID-ELECTIVE-38 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

CPU: 18ms
CGroup: /system.slice/ssh.service
        └─1395 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 02 06:25:41 rojo-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 02 06:25:41 rojo-server sshd[1395]: Server listening on 0.0.0.0 port 22.
Sep 02 06:25:41 rojo-server sshd[1395]: Server listening on :: port 22.
Sep 02 06:25:41 rojo-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
joaquim@rojo-server:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/ssh.service → /usr/lib/systemd/system/ssh.service'.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service'.
joaquim@rojo-server:~$ sudo systemctl start ssh
Failed to start ssh.service: Unit ssh.service not found.
joaquim@rojo-server:~$ sudo apt update
Hit:1 http://ph.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
55 packages can be upgraded. Run 'apt list --upgradable' to see them.
joaquim@rojo-server:~$ sudo apt install openssh-server-y
Error: Unable to locate package openssh-server-y
joaquim@rojo-server:~$
apt install ssh start sudo systemctl update
enable openssh-server-y ssh status
joaquim@rojo-server:~$ sudo apt install openssh-server -y
Error: Unable to locate package openssh-server
joaquim@rojo-server:~$
apt install openssh-server-y ssh status systemctl -y
enable openssh-server ssh start sudo update
joaquim@rojo-server:~$
apt install openssh-server-y ssh status systemctl -y
enable openssh-server ssh start sudo update
joaquim@rojo-server:~$ sudo systemctl start ssh
joaquim@rojo-server:~$ sudo systemctl start ssh
joaquim@rojo-server:~$ sudo apt install openssh-server -y
openssh-server is already the newest version (1:10.0p1-2ubuntu3.5).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14 linux-image-unsigned-7.0.0-14-generic linux-modules-7.0.0-14-generic linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic linux-main-modules-zfs-7.0.0-14-generic linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 55
joaquim@rojo-server:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
joaquim@rojo-server:~$ sudo systemctl start ssh
joaquim@rojo-server:~$
```

Check the status of your server, update and download SSH

```
SOIO-ELECTIVE-38 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Ubuntu 26.04 LTS rojoserver tty1
rojoserver login: joaquin
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

+ Documentation:  https://docs.ubuntu.com
+ Management:    https://landscape.canonical.com
+ Support:        https://ubuntu.com/pro

System information as of Wed Sep  2 07:30:36 AM UTC 2026

System load: 0.06           Memory usage: 2%    Processes:   161
Usage of /:  15.0% of 47.93GB Swap usage:   0%    Users logged in: 0

+ Canonical Workshop gives developers fast, composable, reproducible, and
  secure developer environments that are perfect for agentic workflows.
  https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

joaquin@rojoserver:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:78:4b:f1 brd ff:ff:ff:ff:ff:ff
    altname enp000c27784bf1
    inet 192.168.56.101/24 metric 100 brd 192.168.56.255 scope global dynamic enp0s3
        valid_lft 600sec preferred_lft 600sec
    inet6 fe80::208:2778:4b:f1/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
joaquin@rojoserver:~$ ufw status
joaquin
joaquin@rojoserver:~$ _
```

Check ip

```
Sep 2 16:00
joaquin@rojoserver: ~ — ssh joaquin@192.168.56.101

joaquin@inigo-l-rojo:~$ ssh rojoserver@192.168.56.101
ssh: connect to host 192.168.56.101 port 22: No route to host
joaquin@inigo-l-rojo:~$ ssh rojoserver@192.168.56.101
ssh: connect to host 192.168.56.101 port 22: No route to host
joaquin@inigo-l-rojo:~$ ssh rojoserver@192.168.56.101
rojoserver@192.168.56.101's password:
Permission denied, please try again.
rojoserver@192.168.56.101's password:
rojoserver@192.168.56.101: Permission denied (publickey,password).
joaquin@inigo-l-rojo:~$ ssh joaquin@192.168.56.101
joaquin@192.168.56.101's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

+ Documentation:  https://docs.ubuntu.com
+ Management:    https://landscape.canonical.com
+ Support:        https://ubuntu.com/pro

System information as of Wed Sep  2 07:35:34 AM UTC 2026

System load: 0.01           Memory usage: 2%    Processes:   146
Usage of /:  15.0% of 47.93GB Swap usage:   0%    Users logged in: 0

+ Canonical Workshop gives developers fast, composable, reproducible, and
  secure developer environments that are perfect for agentic workflows.
  https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

joaquin@rojoserver:~$
```

Connect by ssh server@ip